Human-In-The-Loop (HITL)

This document defines the **four gates** where a human's decision moves an artifact forward, the **structural refusals** that make those gates unskippable, and what runs **without supervision**.

The four HITL gates

Each gate is a point where the pipeline pauses until the human acts. The first builds the ontology; the others promote artifacts up the lifecycle.

Gate 1 — Define the layer ontology

- **When:** after the documents have been seeded, via `rag define-layers`
- **What the human does:**
- Names each layer (e.g. `mechanical`, `energy`, `signals`, `operational`, `reference`)
- Assigns a short URN prefix and a one-sentence description per layer
- Optionally lists classification keywords
- **Outcome:** the ontology is written once and frozen for the project's lifetime
- **If skipped:** no other command runs — every downstream step reads this ontology

Gate 2 — Approve a layer per document

- **When:** after the inventory has been seeded, via `rag propose-layers`
- **What the human does:** for each PDF, see the layer the system proposed and:
- **Approve** — keeps the proposal
- **Pick a layer letter** — overrides with a different layer
- **Skip / Quit** — defers the decision

- **Outcome:** every document carries a confirmed layer
- **If skipped:** URN minting and every later step refuses to run

Gate 3 — Review proposed regions

- **When:** after the verifier and the re-crop agent have finished their automatic rounds, via `rag review`
- **What the human does:** for each proposed crop on a page:
- **Approve** — region is locked in, eligible for captioning
- **Enlarge bbox** — scale the bounding box by a factor; the crop is re-rendered
- **Reject** — region is labeled as REJECTED in the database (HITL_approval=false) and its on-disk checklist item is permanently marked as REJECTED to prevent re-drafting. Symmetrically, the physical crop files are preserved to maintain coordinate and relational mappings for any child detail views, while safely excluding the rejected region from final catalog publications.
- **Skip / Quit** — defers the decision
- **Outcome:** only approved regions move into captioning, URN minting, and indexing
- **If skipped:** unapproved regions never receive a URN

The rejection workflow behaves in a unified, non-destructive manner for both macro and micro regions. When a region is rejected by a human reviewer, the system does not physically delete the record or unlink its crop file. Instead, the region is labeled with a rejected status in the database, and its approval flag is set to false. Symmetrically, this preserves the parent-child relational tree and protects coordinate mappings for any child detail views, completely preventing broken database references. Furthermore, the rejected state is immediately written back to the on-disk checklist partition, marking the checklist item as permanently rejected. This ensures that subsequent runs of the visual discovery agents recognize the item as no longer pending, completely preventing infinite drafting loops while ensuring absolute pipeline coordination.

Gate 4 — Review assets

- **When:** after every asset has been assigned a layer automatically, via `rag review --assets`
- **What the human does:** for each asset, see the crop, caption, vision description, and assigned layer, then:
- **Pick a layer letter** — sets the layer (overriding the automatic pick if needed) and approves in one keystroke

- **Approve** — keeps the assigned layer (refused while the layer is still `unknown`)
- **Reject** — removes the asset from the inventory
- **Skip / Quit** — defers the decision
- **Outcome:** every approved asset becomes citable and is included in the exported catalog
- **If skipped:** the asset stays in the inventory but is silently excluded from the published catalog

What runs without the human

These steps are deterministic or judgement-free; human review would add no signal.

- File hashing and PDF metadata extraction
- Page rendering, grid overlays, crop arithmetic
- Per-page language detection
- Region proposing by the cropping agent
- Crop quality verification and automated re-cropping
- Caption and vision-description generation
- Per-asset layer assignment by the vision model
- Vector and full-text index building
- Query routing, retrieval, and answer synthesis
- A warmup query at the end of the pipeline that loads the embedding models into memory, so the first real question afterwards runs without setup delay

The structural refusals

Three points where the system refuses to proceed until the right gate has been cleared. These are what make HITL load-bearing rather than aspirational.

1. **URN minting** refuses to start while any document is unapproved (Gate 2 not cleared)

2. **Index building** refuses to start while any document or asset is unapproved (Gates 2 and 4 not cleared)
3. **Export** emits only approved assets — unapproved entries are silently dropped from the catalog, but the export summary reports the count so nothing vanishes unnoticed

The two-flag promotion ladder

Every asset climbs three stages from minted to citable.

```
Stage 1 — Minted               Stage 2 — Layer assigned      Stage 3 — Human approved

needs_verification: true   →   needs_verification: false  →   needs_verification: false
HITL_approval:      false      HITL_approval:      false      HITL_approval:      true
layer:              unknown    layer:              <name>     layer:              <final>

not citable, not exported      not citable, not exported      citable, in catalog
```

Two flags drive the ladder:

- `needs_verification` — records whether the automated step has run; informational only
- `HITL_approval` — records the human's decision; the **only** flag that gates export

The URN is minted at Stage 1 and never changes. The layer is a free field on the asset, not part of the identifier — so a layer change at Gate 4 leaves the URN untouched.

The same two-flag pattern applies to documents (Gates 1 → 2) and to regions (Gates 3 → captioning → URN minting).

Drift detection

The automated integrity check reports issues directly to the terminal. The human reads the report, fixes the underlying cause, and re-runs.

Issue	Meaning	Recovery
Hash mismatch	A source PDF was edited or replaced	Re-seed documents and re-walk the gates that touched the changed file
Missing crop	A region's crop file is gone from disk	Re-run region verification with reset; the system re-cuts the missing crop
Broken region link	An asset points at a missing or unapproved region	Re-run region review or URN minting
Flag summary	Counts of computer-verified vs human-approved assets	Informational — only the approval flag gates export

Where each command sits

Command	Role in the HITL story
`rag define-layers`	**Is** Gate 1 (ontology definition)
`rag propose-layers`	**Is** Gate 2 (document layer approval)
`rag review`	**Is** Gate 3 (regions review pass)
`rag review --assets`	**Is** Gate 4 (assets layer review pass)